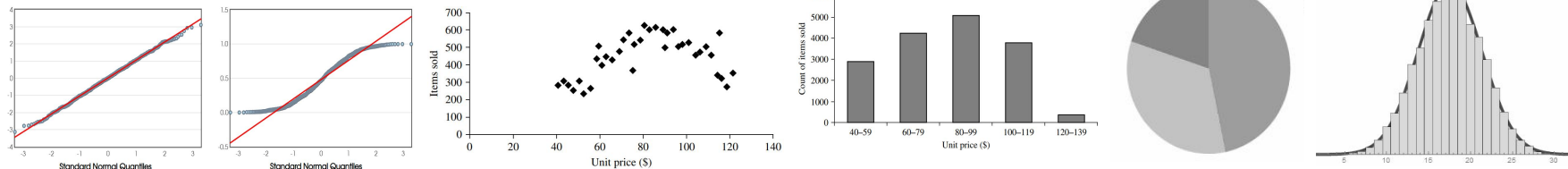


Data Science

CSE558

Overview

- Plots



- Probability Theory

- Basics
- Random Variables
 - R.V. x such that $E[x] = \infty$
 - Linearity of Expectation on Dependent Variables?
- Bernoulli ($E[X] = p$)
- Binomial ($E[X] = np$)
- Geometric ($E[X] = 1/p$)

Overview

- Probability Theory
- Hypothesis Testing

Bernoulli Random Variable

Let you run an experiment that succeed with probability p and fails with probability $1-p$. A Bernoulli random variable X measures success or failure of the experiment,

$$X = \begin{cases} 1, & \text{if the experiment succeeds} \\ 0, & \text{otherwise} \end{cases}$$

The random variable X is also called indicator random variable,

The expectation of the random variable is $E[X] = p$.

One event with success and failure!

Binomial Random Variable

A binomial random variable $\mathbf{X}$ with parameters $\mathbf{n}$ and $\mathbf{p}$, is denoted by $\mathbf{B}(\mathbf{n}, \mathbf{p})$. It takes non negative integer values and defines the probability distribution on $\mathbf{i} = 0, 1, \dots, \mathbf{n}$.

$$\mathbf{P}(\mathbf{X} == i) = \binom{n}{i} p^i \cdot (1 - p)^{n-i}$$

Multiple events with success & failure!

It is aggregation of Bernoulli random variables.

Geometric Random Variable

A geometric random variable $\mathbf{X}$ with parameter $\mathbf{p}$ takes positive integer values. The probability distribution on $\mathbf{i} = 1, 2, \dots$ is,

$$\mathbf{P}(\mathbf{X} == i) = (1 - p)^{i-1} \cdot p$$

$$E[X] = 1/p.$$

Coupon Collector Problem

Suppose there is bag with k types of coupons each having infinitely many copies. If we collect one coupon uniformly at random from the bag, then how many coupons we have to collect until we see at least one coupon of every type.

The event of observing the i^{th} unique coupon follows a geometric distribution. Its probability of success is $p_i = \frac{k - (i - 1)}{k}$. Consider the random variable X_i

$$\mathbf{X}_i = \left\{ j, \quad \text{with probability } (1 - p_i)^{j-1} p_i \right.$$

So, $E[X_i] = 1/p_i$ (#coupons needed to see i^{th} unique coupon).

Coupon Collector Problem

Suppose there is bag with k types of coupons each having infinitely many copies. If we collect one coupon uniformly at random from the bag, then how many coupons we have to collect until we see at least one coupon of every type.

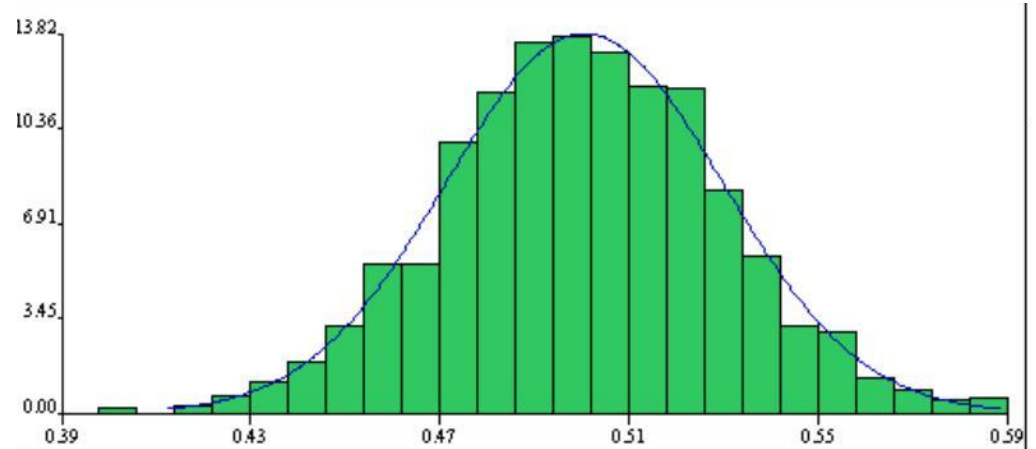
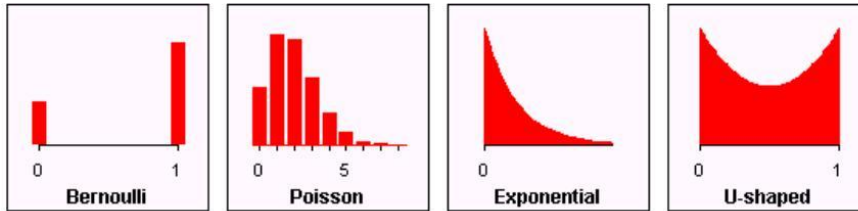
$\mathbf{E}[\mathbf{X}_i] = 1/p_i$ (#coupons needed to see i^{th} unique coupon).

So, #coupons you collect to see at least one coupon of each type is

$$\mathbf{E} \left[\sum_{i=1}^k \mathbf{X}_i \right] = \sum_{i=1}^k \mathbf{E}[\mathbf{X}_i] = \sum_{i=1}^k \frac{1}{p_i} = \sum_{i=1}^k \frac{k}{k-i+1} = k \log(k)$$

Central Limit Theorem

The distribution of the sum of a large sample tends to follow a normal distribution.



Overview

- Probability Theory
- Hypothesis Testing

Game

Let your friend claim on an average she can correctly guess the suit of a randomly picked card with more than 0.25 probability.

To test the correctness of her claim, you randomly picked 200 cards and she correctly guessed more than 52 of them.

Do you believe her claim?

Game

Let your friend claim on an average she can correctly guess the suit of a randomly picked card with more than 0.25 probability.

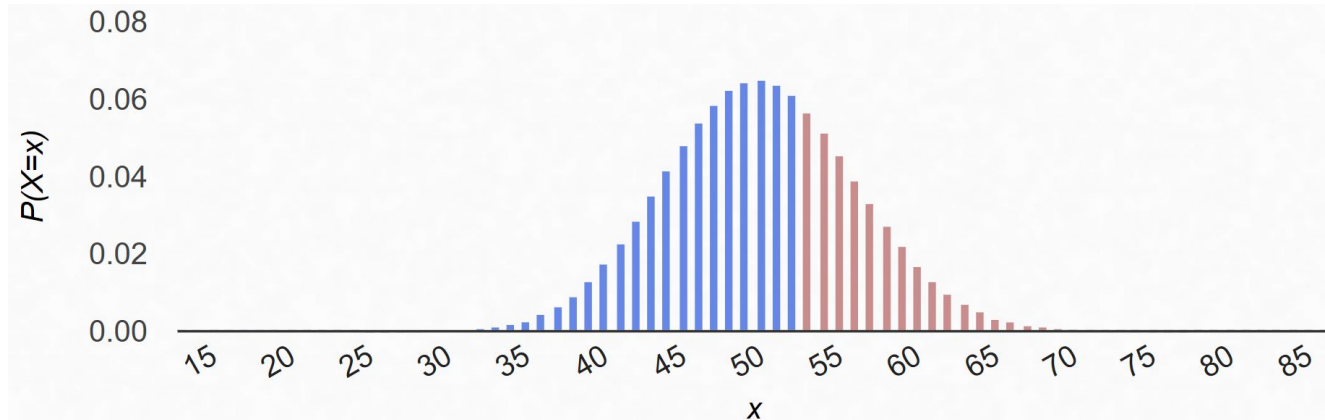
To test the correctness of her claim, you randomly picked 100 cards and she correctly guessed more than 27 of them.

Do you still believe her claim?

Analyze

If the guesses are random then with probability $\frac{1}{4}$ they would be correct.

So with $n = 200$ and $p = \frac{1}{4}$, plot the histogram of the binomial distribution.

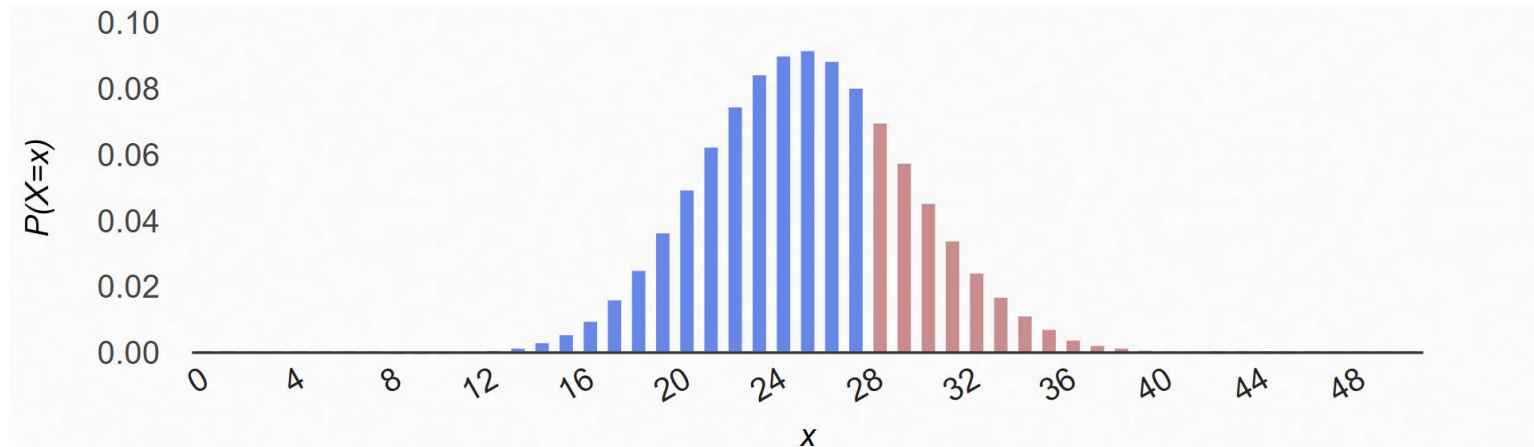


$$P(X \geq 53) = 0.337.$$

Analyze

If the guesses are random then with probability $\frac{1}{4}$ they would be correct.

So with $n = 100$ and $p = \frac{1}{4}$, plot the histogram of the binomial distribution.

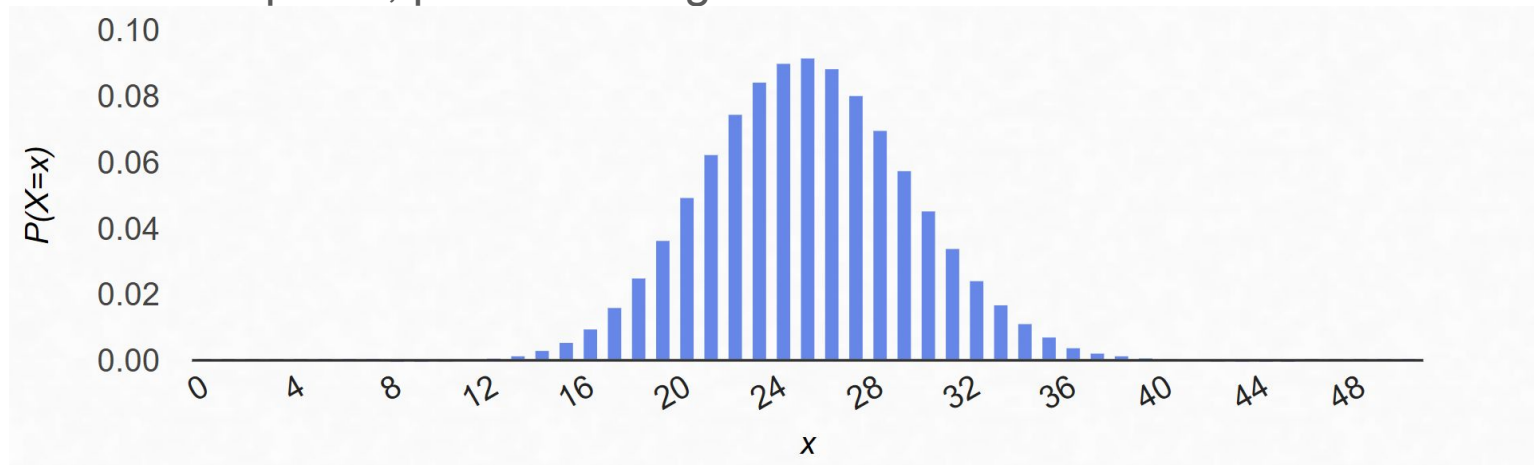


$$P(X \geq 28) = 0.278.$$

Analyze

Lets say your friend got more than 42 correct out of 100!

Again, if the guesses are random then with probability $\frac{1}{4}$ they would be correct. So with $n = 100$ and $p = \frac{1}{4}$, plot the histogram of the binomial distribution.



$P(X \geq 43) = 0.00003$. It is 1 in 33,333 trials!

Parameter Estimation

Medical science in the west is superior than that in India's.
Let 31% of sample agrees to it.

- Empirical Mean: Mean from the sample.
0.31
- Confidence level: Region containing (most believable) parameter
95% confidence level $\Rightarrow$ 5% level of significance = 1.96
- *margin of error*: Ignored region. Here its $1.96 \times \sigma(\text{sample})$
Confidence interval: $0.31 \pm 1.96 \times 0.01 = (0.29, 0.33)$

In Practice: 95% confidence interval or 5% margin of error / level of significance.

Limited Samples

Large samples are scarce due to financial or ethical constraints.

- New-car-assessment-program (NCAP)
- Adverse effect of medicine or vaccine
- Interviews or surveys

Estimating Sample Size

- Confidence Interval
- Margin of error

Compute the sample size for the event where a 95% confidence interval for the population proportion has margin of error equal to 0.1?

In practice: at least 15 success and 15 failure samples.

Hypothesis Testing

Assumption: Data generation, distribution etc.

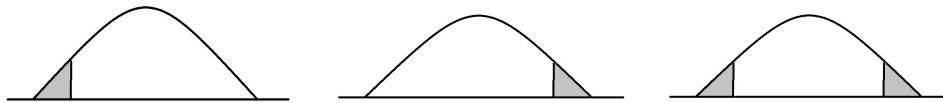
Hypothesis:

- Null Hypothesis (H_0): Status quo, i.e., declared true, until it is tested false.
- Alternate Hypothesis (H_1): Research hypothesis, i.e., range of situation that is to be diagnosed.

Statistics (T): A function that measures distance between empirical estimate and null hypothesis value.

P-value: Probability that measures whether test statistics inclined towards H_0 or H_1

Critical Region: One sided & Two sided.



Reject H_0 based on Critical Value